CAPSTONE 64: AI-Assisted Exploratory Data Analysis

Final Project Report

Project question: Can AI speed up EDA on a public dataset without introducing errors?

Dataset: Student Mental Health During Online Learning — 1,000 records and 10 columns.

1. Objective

The project uses AI assistance to plan and perform exploratory data analysis (EDA), execute analysis in a Jupyter notebook, verify numerical outputs, visualize patterns, and summarize findings. The analysis focuses on stress level, exam anxiety, reported academic-performance change, and lifestyle measurements.

2. Workflow

1) Load and inspect the dataset. 2) Check missing values and duplicate rows. 3) Calculate counts and percentages. 4) Compare stress groups with exam anxiety, academic-performance change, screen time, sleep duration, and physical activity. 5) Calculate correlations between numerical variables. 6) Create charts using Matplotlib and Seaborn. 7) Review AI-assisted interpretations against notebook outputs.

3. Data quality

The supplied notebook screenshots show 1,000 rows and 10 columns, with zero missing values reported for the inspected columns and zero duplicate rows reported.

4. Main findings

Stress level distribution

Stress level	Students	Percentage
Medium	492	49.2%
Low	327	32.7%
High	181	18.1%

Exam anxiety

Response	Students	Percentage
Yes	513	51.3%
No	487	48.7%

Academic performance change

Reported change	Students	Percentage
Same	399	39.9%
Improved	303	30.3%
Declined	298	29.8%

Lifestyle averages by stress level

Stress level	Screen time (h/day)	Sleep (h)	Physical activity (h/week)
High	7.06	6.47	5.15

Low	6.88	6.51	4.84
Medium	6.87	6.40	5.08

5. Relationship analysis

Stress level vs exam anxiety (counts)

Stress level	Anxiety: No	Anxiety: Yes	Total
High	81	100	181
Low	166	161	327
Medium	240	252	492

Stress level vs academic-performance change (counts)

Stress level	Declined	Improved	Same	Total
High	51	54	76	181
Low	92	104	131	327
Medium	155	145	192	492

6. Numerical correlation

The displayed pairwise correlations among age, screen time, sleep duration, and physical activity are close to zero (from -0.03 to 0.03). In this dataset, these numerical variables show very weak linear relationships with one another.

7. Visualizations

The notebook includes a stress-level distribution, exam-anxiety distribution, exam anxiety by stress level (stacked percentage bar chart), academic-performance change by stress level, and boxplots comparing screen time, sleep duration, and physical activity across stress groups.

8. AI use and verification

AI assistance was used to support EDA planning, code drafting, interpretation, and report organization. Numerical findings in this report are taken from the supplied notebook screenshots. Interpretations are kept descriptive and are not treated as evidence of causation.

9. Conclusion

Medium stress was the most common reported category (49.2%). A slight majority of students reported anxiety before exams (51.3%). Reported academic performance was unchanged for 39.9%, improved for 30.3%, and declined for 29.8%. Group averages for screen time, sleep, and physical activity were relatively similar. The displayed correlations between the numerical lifestyle variables and age were negligible. These results describe this dataset and should not be generalized beyond it without further validation.

10. Limitations

This is an observational, descriptive EDA. The results do not establish causation. Conclusions are limited to the fields and outputs shown in the notebook evidence.